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Heat Transfer and Film Cooling Following Injection Through Inclined Circular Tubes

Film cooling effectiveness and heat transfer are measured downstream of injection through discrete holes into a turbulent mainstream boundary layer. Air is injected through both a single hole and a row of holes spaced at three-diameter intervals and inclined at an angle of 35 deg to the main flow. There is little difference between the heat transfer coefficient with blowing and without blowing at low blowing rates (mass flux ratios). In fact, at low blowing rates, injection is found to decrease somewhat the heat transfer coefficient from that measured without blowing. As the mass flux ratio increases past unity, the heat transfer coefficient increases especially with injection through a row of holes. The peak heat transfer is usually found at the edge of the spreading jets (i.e., between two holes). At a blowing rate near two, the lateral average of the heat transfer is as much as 27 percent higher than the heat transfer with no blowing. The increase in heat transfer is attributed to the interaction between the jets and the free stream, causing high levels of turbulence.

Introduction

With film cooling, a coolant is ejected locally through the wall of a structure in such a way that it creates a film along the surface, thereby protecting the structure from exposure to a hot gas stream. In a typical film cooling application, the problem is to predict or measure the relationship between the wall temperature distribution and the heat transfer. The geometry and mainstream and secondary flows may be fixed or may be permitted to vary. One may require the wall temperature for a given set of conditions or optimize the geometry, mainstream or coolant flow while maintaining the wall temperature below some critical value.

The heat transfer coefficient with film cooling is defined as

$$h = \frac{q}{T_w - T_{aw}}. \tag{1}$$

The adiabatic wall temperature is often presented in dimensionless form as the film cooling effectiveness. For low speed, constant property flow the film cooling effectiveness is given by

$$\eta = \frac{T_{aw} - T_r}{T_2 - T_r}. \tag{2}$$

The heat transfer coefficient, as defined by equation (1), is often relatively close to the value without injection. The adiabatic wall temperature, however, can vary considerably and is more difficult to predict. Most film cooling studies are therefore concerned with the determination of the film cooling effectiveness.

A large number of parameters are involved in film cooling. The shape of the channel through which the coolant flows and the angle of injection can be altered for both two-dimensional and three-dimensional film cooling. In addition, the spacing between holes and the number of rows of holes can be varied for the three-dimensional case. The film cooling process also depends on the dimensionless parameters describing the main flow as well as the coolant flow. The ratios of the velocities and densities of the flows are especially important. These quantities are often grouped into the parameters known as the blowing rate (mass flux ratio) and momentum flux ratio. The blowing rate is expressed as

$$M = \frac{\rho_2 U_2}{\rho_\infty U_\infty} \tag{3}$$

and the momentum flux ratio is given by

$$I = \frac{\rho_2 U_2^2}{\rho_\infty U_\infty^2} \tag{4}$$

The subscript 2 denotes the coolant flow; and the subscript ∞ represents the main flow. Other important parameters are the Reynolds number of the main flow, the turbulence in the main flow and the thickness of the mainstream boundary layer at the point of injection. If large temperature differences are employed, the variation of properties throughout the flow field is important.

Two-dimensional film cooling has been studied rather extensively [1],[1] the bulk of the work being concerned with the determination of the film cooling effectiveness. Heat transfer studies

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[1] Numbers in brackets designate References at end of paper.

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with two-dimensional film cooling indicate that injection has little effect on the heat transfer coefficient for blowing rates less than 0.5. At higher blowing rates, the heat transfer coefficient near the slot is usually observed to be somewhat higher than without injection. Its value in this region seems to be quite dependent on injection geometry. Further downstream, the heat transfer coefficient approaches the value without injection.

Three-dimensional film cooling has not been studied as extensively as two-dimensional film cooling. Most of the earlier research was concerned with determination of the film cooling effectiveness [2-13] and is reviewed in [1, 14, and 15].

Information concerning heat exchange between the wall and gas flow in a three-dimensional film cooling environment is sparse. Metzger and co-workers [16, 17] measured the average heat transfer for injection through a row of holes at angles of 20 and 60 deg and hole spacings of 1.55 and 1.71 diam. Results for injection at 20 deg do not differ significantly from results without injection. For 60-deg injection the average heat transfer coefficient was higher than that without injection in the region immediately downstream of the holes but soon decreased to the same value. Burggraf and Huffmeier [19] measured average heat transfer downstream of injection through single and staggered double rows of holes at an angle of 35 deg. For blowing rates less than 1.0, they did not find significant differences between results with and without injection. At blowing rates of 1.5 and 2.0, the heat transfer was greater than occurs without injection. The heat transfer near injection at these higher blowing rates is correlated by the equation for heat transfer without injection when the mass velocity is replaced with the coolant mass velocity.

In the present investigation local values of the heat transfer coefficient are determined for injection of heated and unheated jets of air through holes. Secondary injection is through a row of tubes spaced at three diameter intervals across the span and inclined at an angle of 35 deg to the main flow (Fig. 1). A few measurements are conducted using a single tube from the row. The outlets of the tubes are flush with the surface on which measurements are conducted. A constant heat flux is generated electrically at the test surface and local surface temperatures are measured. Experiments are conducted at moderate temperatures. Differences in the density ratio and variation of fluid properties throughout the flow field should be taken into consideration when using the results for high temperature applications.

Apparatus

The experiments are conducted in a subsonic, open-circuit wind tunnel. The air mainstream in the tunnel flows from the room through an entrance section, the test section, a diffuser, a blower, and finally through a silencer before being discharged

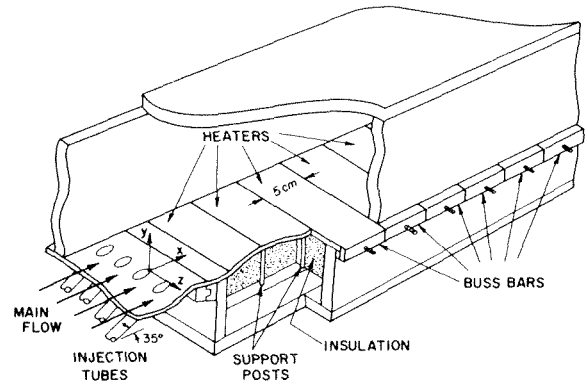

Fig. 1 Test section

outside the building. The secondary, or injected air, is supplied by the building air compressor. The flow rate is controlled by a pressure regulator and needle valve and is measured with a thin plate orifice meter. Temperature fluctuations introduced by the compressor are eliminated by passing the air through a long coiled copper tubing submerged in a large tank of water. The air is heated in a stainless steel tube around which heating tapes are wrapped. The heated air flows into a plenum chamber that provides uniform flow to the injection tubes. This system is described in greater detail elsewhere [5, 6, 9].

The test section is 20.3 cm by 20.3 cm in cross section, and 153.8 cm long. The bottom wall is constructed of Textolite (linen based phenolic manufactured by General Electric) and the top and side walls are constructed of Plexiglas and Textolite. The test section consists of three segments.

The first segment of the test section contains an impact probe and wall pressure tap to determine the free stream velocity, a thermocouple probe to measure the temperature of the main flow, and two thermocouple junctions embedded in the bottom wall to determine the free stream recovery temperature. The bottom wall is thin (0.32 cm thick) to allow it to respond quickly to any temperature changes in the main flow. A 0.064 cm dia boundary layer trip wire is located on the bottom wall approximately 3.8 cm downstream of the contraction section. An additional sandpaper-type trip is located on the bottom surface on the contraction section about 24 cm upstream of its outlet. This additional trip is included to provide a thicker boundary layer than was used in previous studies [5-10].

The next segment contains a row of five tubes at an angle of 35 deg to the direction of main flow as shown on Fig. 1. The 1.18 cm ID tubes, spaced at three-dia intervals, are cemented into the

Nomenclature

D = diameter of injection tube
h = heat transfer coefficient defined using difference between wall and adiabatic wall temperature, see equation (1)
$\bar{h}$ = heat transfer coefficient defined using difference between laterally averaged wall and adiabatic wall temperature, see equation (5)
h_0 = heat transfer coefficient without injection
I = momentum flux or dynamic pressure ratio, $\rho_2 U_2^2/\rho_\infty U_\infty^2$
M = blowing rate, $\rho_2 U_2/\rho_\infty U_\infty$
q = wall heat flux
$\bar{q}$ = laterally averaged wall heat flux
Re_D = Reynolds number using free stream velocity and injection tube diameter ($\mathrm{Re}_D = \rho_\infty U_\infty D/\mu_\infty$)
S = slot width for two-dimensional film cooling and equivalent slot width for three-dimensional film cooling, $S = \pi D/$(4 times center-to-center hole spacing)
T_{aw} = adiabatic wall temperature
$\overline{T_{aw}}$ = laterally averaged adiabatic wall temperature
T_r = mainstream recovery temperature
T_w = wall temperature
$\overline{T_w}$ = laterally averaged wall temperature
T_2 = injection temperature
T = mainstream temperature
U_2 = mean velocity in injection tube
U_∞ = mainstream velocity
X = distance downstream of downstream edge of injection hole, see Fig. 1
Y = distance normal to test surface, see Fig. 1
Z = lateral distance from center of injection hole, see Fig. 1
δ^* = boundary layer displacement thickness
δ_0^* = boundary layer displacement thickness at the upstream edge of the injection hole
η = film cooling effectiveness, see equation (2)
μ_∞ = dynamic viscosity of mainstream gas
ρ_2 = density of injected gas
ρ_∞ = density of mainstream gas

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bottom plate of the segment and ground flush to the tunnel surface. In order to reduce heat conduction from the tube to the plate, the injection plate is thinned from the back side to a thickness of 0.16 cm in the region surrounding the tubes. Thermocouples are soldered on the outside of the injection tubes. The inlet ends of the injection tubes join the plenum chamber mentioned in the foregoing. All tubes are surrounded with fiberglass and styrofoam insulation.

The assembly consisting of the bottom plate of the injection segment, the tubes, tunnel floor, and the plenum chamber is free to slide in the lateral direction. The holes through which the secondary gas flows can thus be located at any lateral position in the tunnel and a single row of thermocouples can be used to measure wall temperature distributions downstream of injection.

The third segment of the test section contains the test plate. Eighteen stainless steel electrical heaters are cemented to the Textolite plate as shown on Fig. 1. The test plate is designed to minimize heat losses out the back and conduction within the wall. Heat should then be transferred to the flow at the location where it is produced and the local heat transfer coefficient can be determined from the local wall temperature. In reality, there are heat losses due to conduction out the back and radiation from the test surface. There are also small errors in the measured local wall temperature due to heat loss through the thermocouple leads and conduction within the wall. Corrections that are applied to the measurements to take these factors into account are discussed in [15]. The 0.16 cm plate that the heaters are cemented to is backed by 5 cm of styrofoam insulation. The stainless steel heaters are 0.025 mm thick by 5.04 cm in the flow direction. They span the entire width of the test plate and are spaced 0.04 mm apart. Copper buss bars are soldered to the ends of the heater strips.

Power for the heaters is provided by two 900 w direct current power supplies wired in series. The current passes through a shunt and through each of the heaters that are all wired in series. Current flow is determined from the voltage drop across the calibrated shunt and the heat generated is calculated from the current flow and the resistance of the heaters. Wall temperatures are measured by 36 gage iron-constantan thermocouples that are embedded in the Textolite plate behind the heater. The thermocouple junction is electrically insulated from the heater.

Operating Conditions and Procedures

Steady-state conditions exist during all tests. For injection of heated jets, a temperature difference of approximately 55 deg C is used, resulting in a density ratio of approximately 0.85. When unheated jets are used, the density ratio is 1.0. The range of variables studied is as follows: free stream velocity $U = 30.5 - 61.0$ m/sec, Reynolds number based on free stream velocity and injection tube diameter $Re_D = 0.22 \times 10^5 - 0.44 \times 10^5$, displacement boundary layer thickness at the point of injection $\delta_0^* = 0.14 - 0.21$ cm, blowing rate $M = 0.1 - 1.95$ and wall heat flux $q = 0 - 0.20$ W/cm^2, resulting in differences between the wall temperature and adiabatic wall temperature in the range 0 − 33 deg C. The wall heat flux is not varied as an independent variable, but only to provide approximately the same temperature difference between the wall and mainstream flow as the free stream velocity is varied.

The secondary air temperature, T_2, is taken as that measured by thermocouples six dia upstream of the tube outlet. The difference between this temperature and the mainstream temperature varies by approximately one percent across the row of holes, most of this variation being between the outside tubes and those adjacent to them. Variation in excess temperature across the inner three tubes is much less than one percent. Velocity at the outlet of the tubes varies by approximately one percent across the row.

Velocity profiles measured at locations both on and off the center line of the test surface and at different positions along the length of the test section without injection are found to be in good agreement with each other and with the fully developed turbulent profile reported by Klebanoff and Diehl [20]. At the point of injection, the boundary layer is approximately 45 percent thicker than in the previous 35 deg injection studies. Boundary layer growth on the walls of the test section causes the main flow velocity to increase by approximately six percent over the length of the test section.

The experimentally determined heat transfer coefficient without secondary injection is used as a reference for tests with injection. The injection holes are covered with thin tape to provide a smooth surface and the wind tunnel is operated at the same velocities as for tests with secondary injection. This heat transfer coefficient depends not only on the free stream velocity and position along the heated wall, but it is also a weak function of the hydrodynamic starting position from where the velocity boundary layer starts to grow to the location where heating of the wall begins. Comparison of the experimental results without injection with theory [21] shows that the theory and results agree quite well at downstream locations where the influence of hydrodynamic starting length is small [15].

The primary flow in the wind tunnel, the secondary flow through the injection tubes and the heat flux from the test surface can be changed to conduct the experiments under a variety of conditions. Adiabatic wall temperatures are determined by operating the tunnel with injection of heated secondary air and no heat flux from the test surface. Heat transfer coefficients are determined with a heated wall and with either heated or unheated injection air. In the former case, the heat transfer coefficient is defined using the difference between the heated wall temperature and the adiabatic wall temperature that was measured under the same flow conditions. With unheated injected air, the heat transfer coefficient is defined using the difference between the heated wall temperature and the free stream recovery temperature. Two separate sets of measurements are necessary to determine the heat transfer coefficient by the first method whereas only one set is necessary for the second. Because these two techniques yield essentially the same results [15], the second method is used for most of the measurements.

Results

Film Cooling Effectiveness. Adiabatic wall temperature measurements for injection through the row of holes display the same trends with X/D, Z/D, and blowing rate as found in previous studies [7, 8] so complete film cooling effectiveness distributions are not included here. Variation of the center-line film cooling effectiveness with Reynolds number and boundary layer thickness at the point of injection is shown on Fig. 2. Included on this figure are results for injection through both a single hole and a row of holes at an angle of 35 deg to the main flow [5–9]. No significant difference has been found between center-line results for injection through this single hole or the row of holes for $M \leq 1.0$ [7, 8]. Data from [5–9] does not permit independent consideration of the variation of film cooling effeviveness with Reynolds number and boundary layer thickness as these two quantities are related. Inclusion of data from the present investigation does permit such a comparison as an additional boundary layer trip is used here. The boundary layer is now thicker at free stream velocities equal to those in [5–9]. The center-line film cooling effectiveness is seen to increase as δ_0^*/D is decreased at a fixed value of the Reynolds number. When the boundary layer at the point of injection is thin, the jet encounters a greater "force" (due to the higher fluid momentum close to the wall) upon leaving the injection tube than for a thicker boundary layer and is turned more, remaining closer to the wall and increasing the film cooling effectiveness. The variation is not as great at the larger values of δ_0^*/D as for the smaller values of δ_0^*/D in [7 and 8]. Fig. 2 also indicates that the film cooling effectiveness seems to increase with the Reynolds number, Re_D, but it is difficult to determine the amount of the increase with the limited amount of data on the figure. If anything, one might expect the opposite effect because of greater mixing at higher Reynolds numbers.

Heat Transfer Coefficient. The heat transfer coefficient downstream of a single 35 deg hole is presented on Fig. 3 for $M = 0.5$, 0.97, 1.46, and 1.95. The heat transfer is within approximate-

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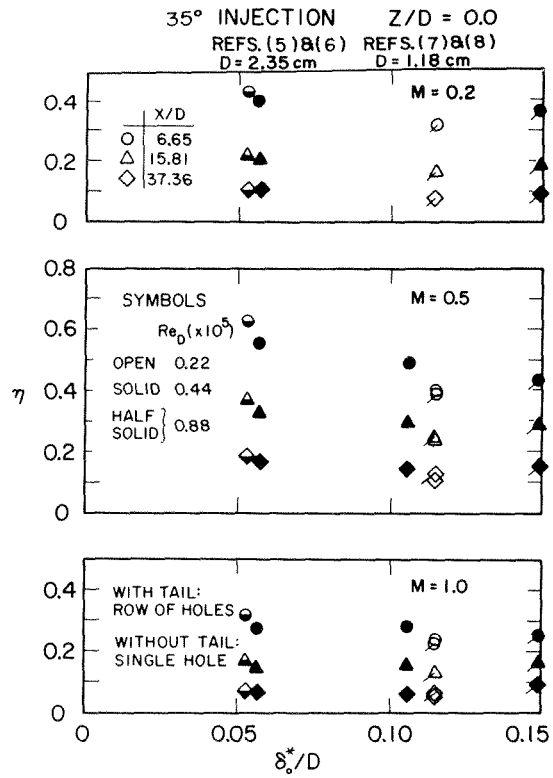

Fig. 2 Variation of center-line film cooling effectiveness with displacement boundary layer thickness

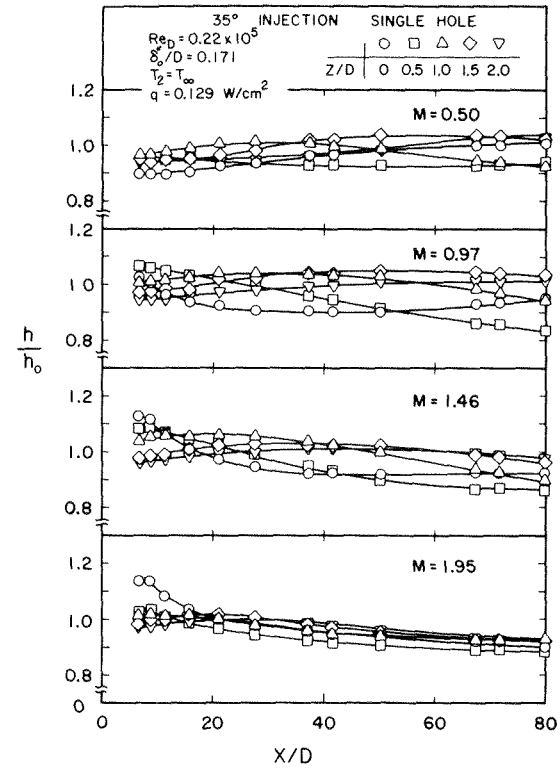

Fig. 3 Heat transfer coefficient downstream of injection through a single hole, $Re_D = 0.22 \times 10^5$

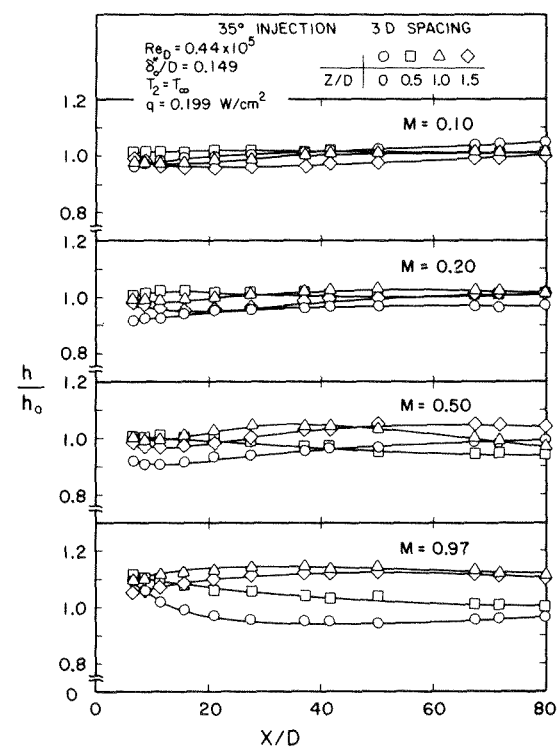

Fig. 4 Heat transfer coefficient downstream of injection through a row of holes, $Re_D = 0.44 \times 10^5$

ly 15 percent of the value without injection for all of these blowing rates. It is often less than that found without injection, especially at the lower blowing rates and far downstream for the higher blowing rates. The injected mass apparently thickens the boundary layer, thus decreasing the heat transfer coefficient in these areas. This effect is countered by increased heat transfer due to turbulence at locations where the edge of the jet and the mainstream interact. The heat transfer coefficient is increased near the hole where the jet and mainflow interact strongly at the higher blowing rates. At the lower blowing rates, interaction between the jet and the mainstream cause the heat transfer coefficient to be higher near the edges of the jet than on the jet center line.

Results from [15] indicate that the heat transfer coefficient for injection through a hole at an angle of 90 deg to the mainflow is approximately 15–20 percent greater than the value without injection at $X/D = 7.33$ and still 10–15 percent greater at $X/D = 35.44$ for $M > 0.5$. Comparison of these values with Fig. 3 shows the heat transfer coefficient for normal injection to be considerably larger than for 35 deg injection. A jet injected at 35 deg has a velocity component in the direction of the main flow; a jet injected normal to the mainstream has no initial velocity component in the direction of main flow. For normal injection there is therefore more interaction between the jet and the mainstream, resulting in higher turbulence levels and a higher heat transfer coefficient than for 35 deg injection. This effect is in agreement with film cooling effectiveness results of [6] where lower effectiveness values and increased spreading of the jet for normal injection when compared to 35 deg injection are attributed to greater interaction between the jet and mainstream during normal injection.

The heat transfer coefficient downstream of injection through a row of 35 deg holes spaced at three dia intervals across the span is presented on Figs. 4 and 5 for Reynolds numbers of 0.44×10^5 and 0.22×10^5, respectively. Except for center-line values near the point of injection, the heat transfer coefficients for $M = 0.1$ and $M = 0.2$ (Fig. 4) fall within 5 percent of the value without injection. For $M = 0.2$, the center-line heat transfer coefficient is somewhat less than the value between holes where the jets interact with the mainstream and with each other. Results for a blowing rate of approximately 0.5 are included on both Fig. 4 and Fig. 5. The heat transfer coefficient is seen to vary from approximately 10 percent lower to 5 percent more than that measured without injection. The center-line heat transfer coefficient is again lower than the value near the edge of the jets. Results at $Re_D = 0.22 \times 10^5$ (Fig. 5) are approximately 3 percent lower than results for the higher Reynolds number.

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Figs. 4 and 5 also include results for a blowing rate of approximately 1.0. For $Re_D = 0.44 \times 10^5$, the heat transfer coefficient on the center line decreases from a value that is approximately 10 percent greater than the flat plate value and nearly constant across the span to a value approximately 4–5 percent below the value without injection for $X/D > 30$. At downstream locations, the heat transfer coefficient between holes where the jets interact with each other and with the main flow is approximately 15–20 percent greater than the center-line value. The heat transfer coefficient at the lower Reynolds number (Fig. 5, $Re_D = 0.22 \times 10^5$) displays similar trends but is approximately 4 percent lower than the heat transfer coefficient for $Re_D = 0.44 \times 10^5$.

Fig. 5 also contains results for blowing rates of $M = 1.45$ and 1.94. The variation of center-line heat transfer coefficient is similar to that observed with $M = 0.99$, decreasing from a value that is nearly constant across the span. The heat transfer coefficient between holes increases rapidly in the downstream direction, apparently due to the jets interacting, reaching a maximum at $X/D = 20$–25 and then decreasing. The maximum value of the heat transfer coefficient (occurring at $Z/D \simeq 1$) is greater than the value without injection by approximately 22 percent for $M = 1.45$ and 37 percent for $M = 1.94$.

Comparison of results for the single hole (Fig. 3) and the row of holes (Fig. 5) shows little difference for $M = 0.5$. At $M = 1.0$, center-line results are similar for the single hole and the row, but heat transfer coefficients between holes are higher for the row than those at the same lateral distance from the single jet. At $M \simeq 1.5$ and 2.0, the heat transfer coefficients following the row of holes are much greater than those for injection through a single hole. Interaction between adjacent jets thus causes increased heat transfer even near the holes at blowing rates above $M = 0.5$.

In film cooling applications, where the surface to be cooled is a good heat conductor, conduction within the wall decreases lateral variations in the wall temperature. The film cooling effectiveness and heat transfer coefficient depend primarily on X/D. A heat transfer coefficient $\bar{h}$ that can be used in these applications is calculated by averaging the wall temperatures that are measured to determine local heat transfer coefficients

$$\bar{h}(X/D) = \frac{\bar{q}}{\bar{T}_w(X/D) - \bar{T}_{aw}(X/D)}. \tag{5}$$

The bar denotes an average in the lateral direction only. For three dia hole spacing,

$$\bar{T}_w(X/D) = \frac{1}{3}\int_{-1.5}^{+1.5} T_w(X/D, Z/D)d(Z/D). \tag{6}$$

The adiabatic wall temperature and approximately constant (variation is due to change of resistance of heaters with temperature) wall heat flux are averaged in the same way.

The heat transfer coefficient defined using laterally averaged wall temperatures is presented on Figs. 6(a) and 6(b) for $Re_D = 0.44 \times 10^5$ and $Re_D = 0.22 \times 10^5$, respectively. Fig. 7 contains a crossplot of this heat transfer coefficient against the blowing rate M. Fig. 6(a) shows that results for $M \leq 0.5$ do not differ from values without injection by more than 3 percent for $Re_D = 0.44 \times 10^5$. The heat transfer coefficient at the lower Reynolds number (Fig. 6(b)) varies from approximately 6 percent below the flat plate value near injection to approximately 3–5 percent below the value without injection at downstream locations for $M = 0.2$ and 0.5. The heat transfer coefficient for heated injection is smaller than for unheated injection by approximately 1 percent for $M = 0.2$ and 1.5 percent for $M = 0.5$. These differences are similar to those that are observed for normal injection through a single tube and are partly due to the higher uncertainties that are associated with heated injection [15].

At $M = 0.99$, the heat transfer coefficient decreases from approximately 10–12 percent above the value without injection near the hole to 5–6 percent above flat plate values at downstream locations for $Re_D = 0.44 \times 10^5$. Fig. 7 shows results for $Re_D = 0.22 \times 10^5$ to be approximately 5 percent lower than those at the higher Reynolds number for $M = 0.99$. The heat transfer coefficient determined with heated injection ($\rho_2/\rho_\infty = 0.85$) is approximately 2 percent higher than that found with unheated injection ($\rho_2/\rho = 1$) at this blowing rate. Since the jet is penetrating into the mainstream for $M = 1$, the increased heat transfer coefficient for heated injection at this blowing rate could be due to the differ-

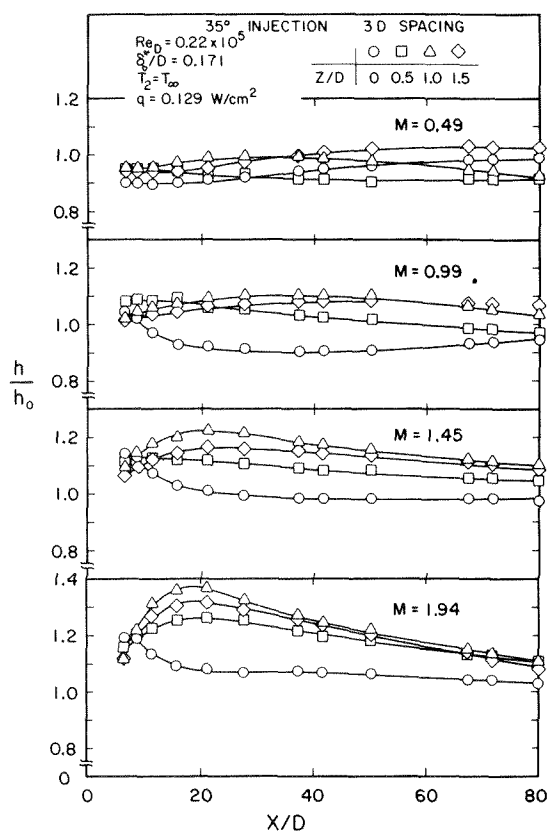

Fig. 5 Heat transfer coefficient downstream of injection through a row of holes, $Re_D = 0.22 \times 10^5$

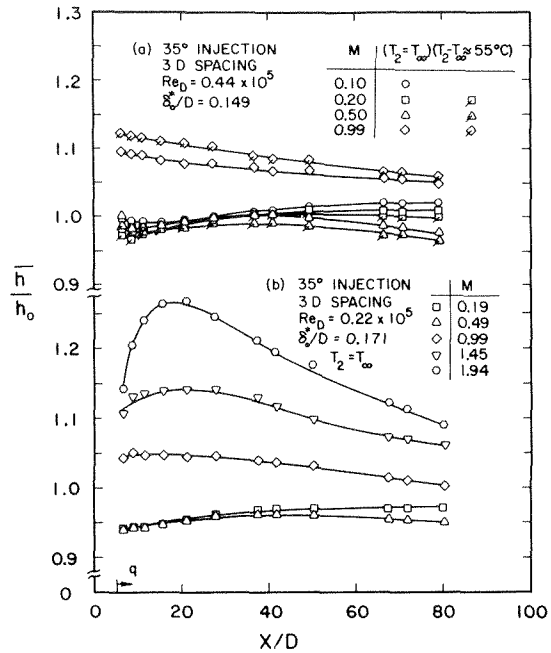

Fig. 6 (a) $RE_D = 0.44 \times 10^5$; (b) $Re_D = 0.22 \times 10^5$. Heat transfer coefficient based on laterally averaged wall temperatures downstream of injection through a row of holes.

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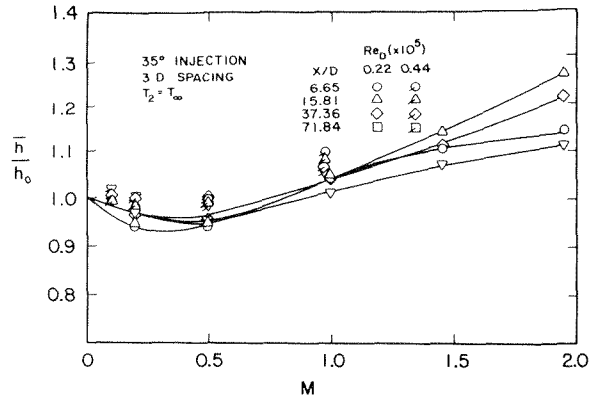

Fig. 7 Variation of heat transfer coefficient based on laterally averaged wall temperatures with blowing rate for injection through a row of holes

ence in the momentum flux ratio I between heated and unheated injection.

For $M = 1.45$ and $M = 1.94$, the laterally averaged heat transfer coefficient increases in the downstream direction, reaches a peak at $X/D \simeq 20$ and then decreases at larger values of X/D. The peak value is about 14 percent greater than the value without injection for $M = 1.45$ and about 27 percent higher for $M = 1.94$. References [7] and [8] show the film cooling effectiveness increasing with X/D for 35 deg injection through a row of holes at $M = 1.5$ and 2. The jets penetrate into the main flow at these blowing rates and the increase in effectiveness is explained as being the result of spreading of the jets. The increased values of the heat transfer coefficient can result from the high turbulence level near the edges of the jets that are spreading toward the wall.

Increased values of the heat transfer coefficient, especially near the point of injection and at the higher blowing rates, have been attributed to turbulence that results from interaction between the jets and the main flow. Turbulence measurements have been reported for normal injection of a single jet into the main flow. References [9 and 10] report the turbulence intensity to be as high as 60 percent in the region near injection for blowing rates greater than 1. Very large eddies have also been observed for normal injection in [15]. Although turbulence levels would not be expected to be as high for 35 deg injection as for normal injection, it is believed that they are still high enough to promote heat transfer.

The heat transfer coefficient based on the average (in both the lateral and downstream directions) wall temperature was determined by Metzger and co-workers [16, 17] for injection through a row of holes at an angle of 60 deg with the main flow. Hole spacings of 1.55 and 1.71 dia were used at blowing rates $M = 0.25$, 0.50, and 0.75. Although the wall temperatures used to calculate $\bar{h}$ in the present work are averaged only in the lateral direction, $\bar{h}$ can be compared with the results of [16 and 17] at low blowing rates where h does not vary much in the downstream direction. The heat transfer reported in [16 and 17] is a few percent above flat plate values at $M = 0.25$, while the results on Figs. 6*(a)* and 6*(b)* for $M = 0.2$ are 3-5 percent below flat plate values at $Re_D = 0.22 \times 10^5$ and approximately equal to the value without injection for $Re_D = 0.44 \times 10^5$. The small difference between the results could be due to the difference in blowing rates, different Reynolds numbers or the different injection geometries. For $M = 0.5$, the heat transfer coefficients of [16 and 17] decrease from approximately 25 percent above the flat plate value near the hole to approximately 10 percent above the value without injection at $X/S \simeq 65$ ($X/S = 65$ corresponds to $X/D = 29.8$ for 1.71 dia spacing and $X/D = 33$ for 1.55 dia spacing). The large difference between these results and the data in Figs. 6*(a)* and 6*(b)* which fall a few percent below the value without injection may be due to differences in geometry or increased mass flow in the mainstream due to secondary injection. The larger injection angle would increase interaction between the jet and mainstream and the smaller hole spacings should increase interaction between neighboring jets. Both of these effects would tend to increase the heat transfer coefficient. For $M = 0.5$, secondary injection apparently increases the mainstream flow rate in [16 and 17] by 5-10 percent [18]. This additional mass flow could increase the heat transfer coefficient by 4-8 percent.

Conclusions

Comparison of film cooling effectiveness for 35 deg injection through a row of holes with results from other investigations indicates that the film cooling effectiveness varies with Reynolds number and boundary layer thickness at the point of injection as well as X/D, Z/D, and M. Fig. 2 shows that the center-line film cooling effectiveness decreases as the boundary layer thickness at the point of injection is increased. When the boundary layer at the point of injection is thin, the jet encounters a greater "force" upon leaving the injection tube than in a thicker boundary layer and is turned faster, remaining closer to the wall and increasing the film cooling effectiveness. This effect, as would be expected, seems most important at $M \simeq 0.5$ which is the blowing rate where penetration of the jet begins to be important. The film cooling effectiveness appears to increase somewhat with Reynolds number Re_D.

The heat transfer coefficient for 35-deg injection through a single hole is observed to be smaller than has been observed for normal injection. The jet injected at 35 deg has a velocity component in the direction of main flow; the jet injected normal to the mainstream has no initial velocity component in the direction of main flow. There is therefore more interaction between the normally injected jet and the mainstream, resulting in higher turbulence levels and a higher heat transfer coefficient than for 35-deg injection. This is in agreement with film cooling results of [6] where lower effectiveness values and increased spreading of the jet for normal injection when compared to 35-deg injection are attributed to greater interaction between the jet and mainstream for normal injection.

There is little difference between the heat transfer coefficient for 35-deg injection through a single hole or through a row of holes spaced at three-diameter intervals across the span for $M = 0.5$. At $M = 1.0$, center-line heat transfer results are similar for the single hole and row, but heat transfer for the row is higher between holes than at the same lateral distance from the single jet. At $M = 1.5$ and 2, heat transfer results for a row of holes are much higher than for injection through a single hole at all lateral positions. Interaction between adjacent jets and between the jets and the mainstream increases heat transfer at blowing rates above $M = 0.5$. A similar effect was observed in [7 and 8] where interaction between jets increased the film cooling effectiveness at blowing rates greater than 1.

At blowing rates $M = 1$, 1.5, and 2, the heat transfer coefficient varies by as much as 20 percent in the lateral direction for 35-deg injection through the row of holes. The heat transfer coefficient is greatest between holes where the jets interact with the mainstream and with each other. At blowing rates $M = 1.5$ and 2, the heat transfer coefficient based on laterally averaged wall temperatures first increases in the downstream direction, reaches a maximum value at $X/D \simeq 20$ and then decreases with X/D. The jets penetrate into the free stream at these blowing rates and then spread toward the wall. The heat transfer coefficient increases with X/D for $X/D \leq 20$ because of the high turbulence level near the edges of the jets that are spreading toward the wall.

The heat transfer coefficient for 35-deg injection through the row of holes at $Re_D = 0.44 \times 10^5$ is approximately 3-5 percent higher than at $Re_D = 0.22 \times 10^5$. The higher heat transfer coefficient at the larger Reynolds number may be due to increased mixing of the secondary flow at the higher Reynolds number and thinner free stream boundary layer.

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